

TAUSIF AHMED

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EDUCATION

Bachelor of Science in Computer Science

Anticipated Graduation: May 2027
Saginaw Valley State University, University Center, MI

PROJECT LEADERSHIP EXPERIENCE

NASA Lunabotics Competition - Systems Engineering Team Lead

Aug 2025 – Present

Saginaw Valley State University

Saginaw, MI

- Lead software, electrical, and mechanical integration for a NASA Lunabotics rover while coordinating requirements, schedules, testing activities, and technical documentation across subsystem teams.
- Developed a Raspberry Pi-based robot software stack using Python, Linux, OpenCV, YOLO, Kinect RGB/depth sensing, TCP sockets, MJPEG streaming, and Arduino serial communication.
- Implemented and debugged real-time teleoperation and autonomy features, including keyboard/Xbox controller input handling, robot command parsing, actuator controls, emergency stop states, telemetry updates, and autonomous mode transitions.
- Built hybrid obstacle-detection logic using object detection and depth data to support autonomous navigation decisions such as forward motion, stopping, turning, recovery behavior, and low-visibility handling.
- Integrated multiple Arduino controllers for left drive, right drive, and linear actuators through serial communication, supporting testing of motor direction, duty-cycle mapping, current behavior, and actuator command reliability.
- Designed and tested a Mission Control interface with live robot video, connection status, bandwidth-aware low-video mode, Arduino/Kinect health indicators, manual lock states, and autonomy status feedback.
- Performed software integration and debugging across Linux, embedded hardware, sensors, network communication, and robot control code to resolve real-time issues during field testing.
- Created NASA-style systems engineering documentation including system requirements, operational procedures, subsystem interfaces, verification planning, design reviews, and project milestone tracking.

RELEVANT WORK EXPERIENCE

IT Support Specialist

June 2025 – Present

Morley Companies, Inc.

Saginaw, MI

- Resolve customer technical issues in a fast-paced support environment while documenting service requests, troubleshooting steps, and solutions clearly and accurately.
- Communicate technical information to users with different levels of experience, strengthening the customer-facing communication skills needed for cross-functional engineering teams.
- Collaborate with team members to maintain smooth daily operations, support high request volume, and improve service quality through organized documentation.

Technical Support Specialist At IT Technical Services

March 2021- January 2025

Saginaw Valley State University

Saginaw, MI

- Provided technical support to 4,000+ students, faculty, and staff by troubleshooting hardware, software, account, classroom technology, and network-related issues.
- Supported setup and deployment of equipment for 2,000+ employees and users, gaining practical experience with hardware configuration, device troubleshooting, and user support workflows.
- Maintained accurate documentation of recurring issues and resolutions to improve support efficiency, knowledge sharing, and future troubleshooting.
- Managed service requests and support appointments using digital ticketing and Microsoft Office tools while maintaining clear communication with end users.

Student Researcher (Accounting, Economics)

October 2023 – February 2024

Saginaw Valley State University

Saginaw, MI

- Collected, cleaned, and analyzed datasets to identify trends and produce reports supporting faculty research objectives.
- Documented findings and communicated results clearly to team members, strengthening data analysis, reporting, and technical communication skills.

Front Desk Support At Career Services

February 2021 – March 2021

Saginaw Valley State University

Saginaw, MI

- Assisted students with resumes, cover letters, job applications, appointment scheduling, and career-event support while maintaining professional communication.

Technical Skills

- **Programming:** Python, C, C++, Java, PHP, JavaScript, Bash/Linux command line
- **Embedded / Robotics:** Raspberry Pi, Arduino, UART/serial communication, USB devices, motor/actuator control, sensor integration, real-time robot command handling
- **Computer Vision & Data:** OpenCV, YOLO object detection, Kinect RGB/depth sensing, depth-based obstacle avoidance, image/data fusion, NumPy
- **Software Tools:** Git/GitHub, Linux, VS Code, Eclipse, Agile-style task tracking, technical documentation, debugging and testing workflows
- **Databases / Web:** MySQL, SQL Server, SQLite, MongoDB, HTML, CSS, JavaScript, AWS basics